

Revive Pragathi Lake

A Sustainable, Self-Funding Ecosystem for Hyderabad

Moving from temporary cleaning to a permanent sustainability loop — transforming a neglected water body into a thriving community asset.



The Challenge: A Space Wasting Away

Stagnant Water & Mosquito Breeding

Still water creates ideal conditions for extreme mosquito breeding, posing a direct public health risk to surrounding communities.

Insufficient Maintenance

The lake is cleaned roughly once per year — a frequency far too low to maintain safe, usable water conditions.

Funding Dependency

Heavy reliance on limited government budgets leads to inconsistent upkeep, foul odors, and degraded environmental quality for residents.



Layer 1: Solar-Powered Anti-Stagnation



How It Works

Floating solar-powered paddles and aerators create constant water circulation, raising dissolved oxygen levels and disrupting the surface tension that mosquito larvae need to survive.



Zero Energy Cost

Fully solar-powered



Natural Larvae Disruption

Continuous surface movement

Layer 2: Nature-Based Larvae Control

Introducing *Gambusia affinis* (Mosquitofish)

We use *Gambusia affinis* because it eats mosquito larvae. That is the primary reason it is the right biological solution for this problem: the fish actively reduces larvae before they can mature into biting mosquitoes.

- Consumes mosquito larvae directly
- Works in shoreline "dead zones" where mechanical aerators cannot reach
- Self-sustaining after initial introduction with no recurring cost



Layer 3: Smart Cleanliness — Barriers & Robots

Passive: Floating Barriers

Trash nets installed at drainage entry points trap garbage before it spreads across the lake surface.

Active: Autonomous Robot

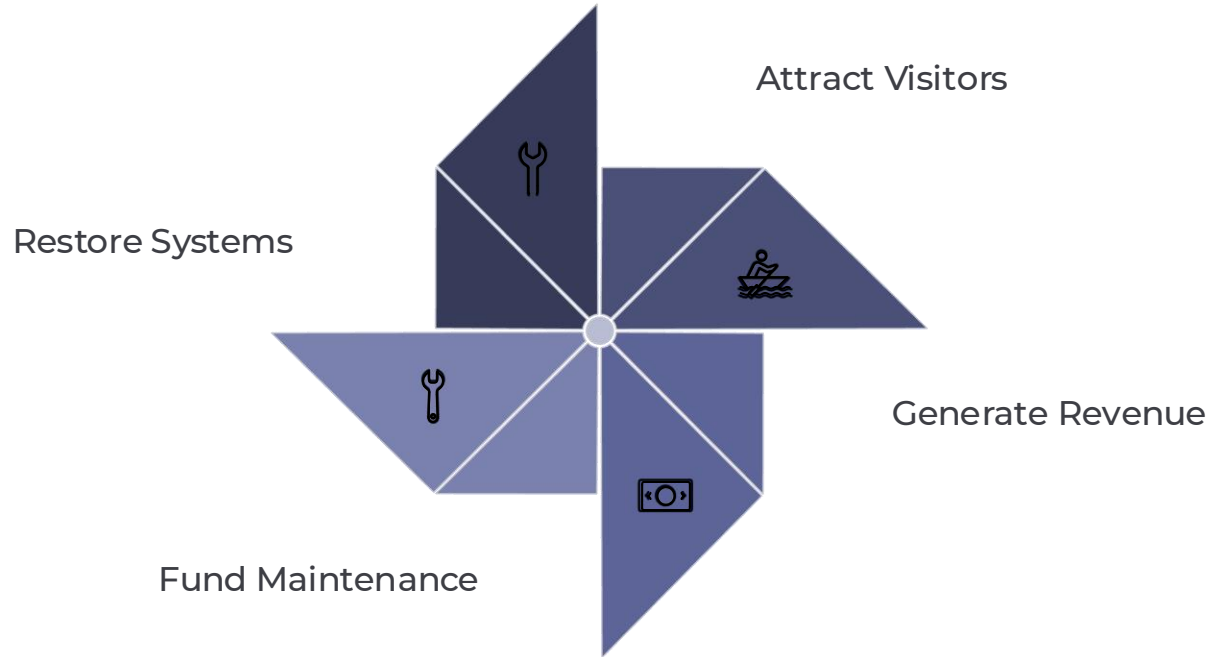
An ESP-32 based, solar-powered robot patrols the surface, collecting floating debris and reducing manual labor needs.

Result: Aesthetic & Functional

Keeps the lake visually appealing for visitors while minimizing operational staffing costs.

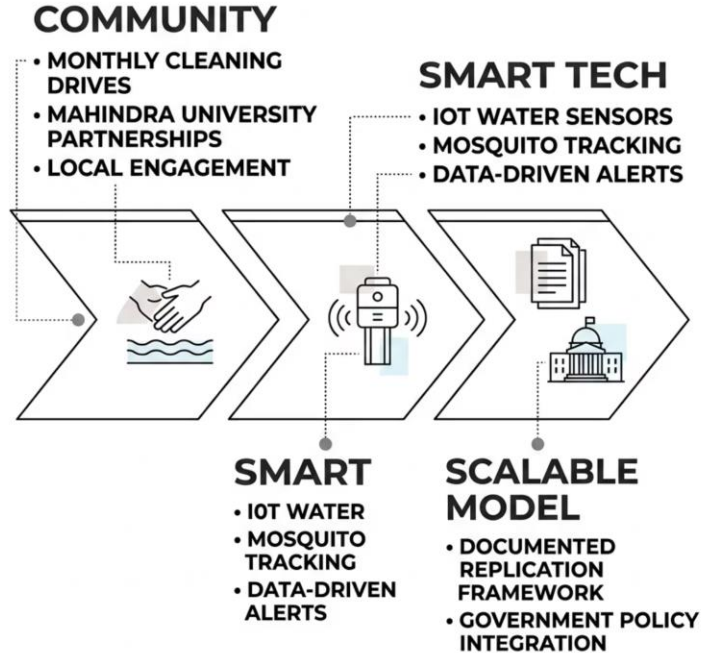


The Sustainability Loop: From Cost Center to Revenue Generator



Cleaner water drives public attraction, which generates revenue — funding the very systems that keep the lake clean. Pragathi Lake becomes a self-sustaining asset, not a recurring budget burden.

Community Ownership & Future Roadmap



A Model Built to Scale

Pragathi Lake is not just a restoration project — it is a blueprint. By combining community participation with smart technology, this model can be replicated across Hyderabad's urban water bodies.

- Student partnerships with institutions like Mahindra University will drive innovation and sustained volunteer engagement.

The Vision: Pragathi Lake Reimagined

3

Integrated Layers

Mechanical, biological, and robotic systems working in harmony

₹0

Net Operational Cost

Revenue from recreation funds all ongoing maintenance

1

Scalable Blueprint

A replicable model for every urban lake in Hyderabad

From polluted and neglected to a vibrant, self-sustaining community hub — Pragathi Lake can lead Hyderabad's urban lake revival.

